Lab 3: Answer Key

Substitutes for Prediction (Quadratic Costs — Canvas Version)

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Parameters: $R = 100$, $r = 20$, $p = 0.7$, $\gamma = 600$, $\kappa = 0.01$, $\lambda = 5$.

$$C(x) = \tfrac{\gamma}{2}x^2 = 300x^2, \qquad c(\Delta) = \tfrac{\kappa}{2}\Delta^2 = 0.005\,\Delta^2$$

$$C'(x) = \gamma x = 600x, \qquad c'(\Delta) = \kappa\Delta = 0.01\,\Delta$$

(A power-law/log-cost variant of this lab is archived as `lab3_answer_key_logcost_VERSION.qmd`. The version below matches the handout `lab3_substitutes_to_prediction_canvas.pdf` that students received.)

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1 Part A: Following a rule with risk management

(a) Expected payoff under $a = 1$ with protection x and insurance Δ:

$$\pi(x, \Delta) = (p + x)R + (1 - p - x)(r + \Delta) - \tfrac{\gamma}{2}x^2 - \tfrac{\kappa}{2}\Delta^2$$

$$= (0.7 + x)(100) + (0.3 - x)(20 + \Delta) - 300x^2 - 0.005\,\Delta^2$$

(b) FOC with respect to x:

$$\frac{\partial \pi}{\partial x} = R - (r + \Delta) - \gamma x = 0 \implies \boxed{R - r - \Delta = \gamma x}$$

That is, $80 - \Delta = 600x$. The marginal benefit of protection is $R - r - \Delta$: an additional unit of protection shifts probability toward the matching state, upgrading the payoff from $r + \Delta$ to R. The marginal cost is γx: cost rises linearly with the protection chosen.

(c) FOC with respect to Δ:

$$\frac{\partial \pi}{\partial \Delta} = (1 - p - x) - \kappa\Delta = 0 \implies \boxed{1 - p - x = \kappa\Delta}$$

That is, $0.3 - x = 0.01\,\Delta$. The marginal benefit of insurance is the probability $1 - p - x$ of a mismatch (when insurance pays off); the marginal cost is $\kappa\Delta$.

(d) From (c): $\Delta = 100(0.3 - x) = 30 - 100x$. Substitute into (b):

$$80 - (30 - 100x) = 600x \implies 50 + 100x = 600x \implies \boxed{x^*_{RULE} = 0.1}$$

$$\Delta^*_{RULE} = 30 - 100(0.1) = \boxed{\Delta^*_{RULE} = 20}$$

(e) Plug back in:

$$\pi^*_{RULE} = (0.8)(100) + (0.2)(40) - 300(0.1)^2 - 0.005(20)^2$$

$$= 80 + 8 - 3 - 2 = \boxed{83}$$

(f) Cross-partial:

$$\frac{\partial^2 \pi}{\partial x\, \partial \Delta} = \frac{\partial}{\partial \Delta}\big[R - (r + \Delta) - \gamma x\big] = -1$$

Since $-1 < 0$, insurance and protection are substitutes: increasing one reduces the marginal benefit of the other. Economically, with more insurance the payoff gap $R - r - \Delta$ that protection closes is smaller, so protection is less valuable — and vice versa.

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2 Part B: A signal that beats the prior but not the rule

Parameters: $e = 0.75$, $\lambda = 5$.

(a) $e = 0.75 > 0.7 = p$. ✓

(b) Under a decision (with $x = 0$):

$$\pi_{DEC}(\Delta) = eR + (1 - e)(r + \Delta) - \tfrac{\kappa}{2}\Delta^2 - \lambda$$

$$= 0.75(100) + 0.25(20 + \Delta) - 0.005\,\Delta^2 - 5 = 75 + 0.25\Delta - 0.005\,\Delta^2$$

(c) FOC: $1 - e = \kappa\Delta \implies 0.25 = 0.01\,\Delta \implies \boxed{\Delta^*_{DEC} = 25}$.

(d) Plug back in:

$$\pi^*_{DEC} = 75 + 0.25(25) - 0.005(625) = 75 + 6.25 - 3.125 = \boxed{78.125}$$

(e) $78.125 < 83$. The firm should **not adopt AI**. Even though the signal beats the raw prior ($e > p$), it cannot beat the rule once the firm has the option to invest in protection and insurance.

(f) $e = 0.75 < 0.8 = p + x^*_{RULE}$. The condition does **not** hold.

The condition $e > p$ says the signal beats the *raw prior*. But the firm hasn't been relying on the raw prior — it invested in protection, effectively boosting its probability of matching the state from $p = 0.7$ to $p + x^*_{RULE} = 0.8$. The signal must beat this *enhanced* prior, not just the raw one. Protection raises the adoption bar for AI.

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3 Part C: A signal that clears the bar

Parameters: $e = 0.85$, $\lambda = 5$.

(a) $e = 0.85 > 0.8 = p + x^*_{RULE}$. ✓

(b) FOC: $1 - e = \kappa\Delta \implies 0.15 = 0.01\,\Delta \implies \boxed{\Delta^*_{DEC} = 15}$.

(c) $\pi^*_{DEC} = 0.85(100) + 0.15(20 + 15) - 0.005(15)^2 - 5 = 85 + 5.25 - 1.125 - 5 = \boxed{84.125}$.

(d) $84.125 > 83$. The firm **adopts AI**. The expected payoff increases by $84.125 - 83 = 1.125$.

(e) $\Delta^*_{DEC} = 15 < 20 = \Delta^*_{RULE}$. Insurance **decreased**. This is consistent with Proposition 4.1: when e is sufficiently high (here $e = 0.85 > p + x^*_{RULE} = 0.8$), the signal is accurate enough that the agent makes fewer errors, reducing the need for insurance against mismatch.

(f) $e = 0.85 > 0.7 = p$. ✓ This is necessary (Corollary 4.1) but not sufficient. Problem 2 showed that $e = 0.75 > p = 0.7$ was not enough — the firm didn't adopt because the signal couldn't beat the protection-enhanced prior $p + x^*_{RULE} = 0.8$.

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4 Part D: A conceptual question

(a) From Problem 2: $\Delta^*_{DEC} = 25 > 20 = \Delta^*_{RULE}$. Insurance rises. ✓

From Problem 3: $\Delta^*_{DEC} = 15 < 20 = \Delta^*_{RULE}$. Insurance falls. ✓

(b) The threshold $e = p + x^*_{RULE}$ governs both the adoption decision and the insurance comparison, and this is not a coincidence. Under a decision, the insurance FOC is $\kappa\Delta = 1 - e$, so $\Delta^*_{DEC} = (1-e)/\kappa$. Under a rule, the insurance FOC is $\kappa\Delta = 1-p-x^*_{RULE}$, so $\Delta^*_{RULE} = (1-p-x^*_{RULE})/\kappa$. Insurance rises iff $\Delta^*_{DEC} > \Delta^*_{RULE}$, i.e., $1 - e > 1 - p - x^*_{RULE}$, i.e., $e < p + x^*_{RULE}$. But the adoption condition requires $e > p + x^*_{RULE}$. The same inequality separates both: $e \gtrless p + x^*_{RULE}$ determines both whether AI is adopted and whether insurance rises or falls.

(c) Proposition 4.1 is comparing two separate optimization problems — "how much insurance would you want *if* you followed a rule?" versus "how much insurance would you want *if* you made a decision?" — not predicting what a freely-optimizing firm would actually do. The comparison is still useful because it tells you what would happen if an agent were forced into a particular regime: regulations, competitive pressure, or organizational mandates can push a firm into making AI-driven decisions at signal qualities it wouldn't have chosen voluntarily. The cross-regime comparison tells us what to expect in those forced cases.

(d) *Example: Criminal sentencing.* Many U.S. jurisdictions adopted mandatory sentencing guidelines (rules) that prescribed fixed sentences based on offense category. More recently, some jurisdictions have required judges to consult risk-assessment algorithms when sentencing — effectively mandating a switch from rules to decisions. If the algorithm's accuracy is mediocre ($e < p+x^*_{RULE}$), the judicial system would be expected to increase insurance-like activities: more robust appeals processes, more frequent parole reviews, sentence-modification hearings, and post-sentencing monitoring. These are all ways to reduce the cost of getting individual decisions wrong — exactly the insurance response Proposition 4.1 predicts when a decision regime is adopted at a low signal quality.

Other acceptable examples include:

- Hospitals mandated to use AI diagnostic tools by regulation, increasing follow-up tests and confirmatory procedures (insurance)
- Content moderation platforms required by the EU Digital Services Act to use automated detection, increasing human review queues and appeals processes (insurance)
- Financial institutions required to use algorithmic credit scoring, increasing manual review for borderline cases (insurance)